Unit 2 Guided Notes — Exponents & Scientific Notation

Same base? Move the exponents. Big or tiny number? Pack it into $a \times 10^n$.

Standards: NC.8.EE.1 · NC.8.EE.3 · NC.8.EE.4

This packet weaves three parts together: **Guided Notes** with worked examples, **Practice** built into each section to try as you learn, and a **Cumulative Review** with a full **Answer Key** at the end. Fill in the blanks and show your work on your printed copy.

Topics Covered: 2.1 Exponent Rules — Product, Quotient, Power, Zero, Negative (NC.8.EE.1) · 2.2 Scientific Notation — Format & Conversions (NC.8.EE.3) · 2.2 Multiplying & Dividing Scientific Notation (NC.8.EE.4) · 2.2 Word Problems with Scientific Notation (NC.8.EE.4)

1 Exponent Rules

↗ **KEY RULE:** Exponent rules only apply when the BASES are the SAME!

Fill the examples and key phrases on your printed copy.

Rule	Formula	Simple Example	Monomial Example	Key Phrase
Product Rule	$x^a \times x^b = x^{a+b}$	________	________	________
Quotient Rule	$x^a \div x^b = x^{a-b}$	________	________	________
Power Rule	$(x^a)^b = x^{a \times b}$	________	________	________
Zero Exponent	$x^0 = 1$	________	________	________
Negative Exponent	$x^{-n} = 1/x^n$	________	________	________

⚠ **PARENTHESES MATTER!**
$-3^2 =$ ________ vs. $(-3)^2 =$ ________
Without ( ), ONLY the base gets the exponent — not the negative sign!

PRODUCT RULE: COUNT THE FACTORS

x x x × x x = x x x x x

x^3 (3 factors) × x^2 (2) = x^5 (5 factors)

$x^3 \times x^2 = x^{3+2} = x^5$ → ADD the exponents

Modeling the Rules — See WHY

*Each rule comes from just writing out the repeated multiplication — **first base**, **second base**, result.*

Product Rule · $x^3 \times x^2 = x^5$

See why → x x x × x x = x x x x x = x^5

Count the factors: 3 + 2 = 5. Same base → ADD the exponents.

Quotient Rule · $x^5 \div x^2 = x^3$

See why → x x x x x ÷ x x → cancel 2 pairs → x x x = x^3

Cancel matching factors: 5 − 2 = 3. Same base → SUBTRACT the exponents.

Power Rule · $(x^2)^3 = x^6$

See why → x^2 x^2 x^2 = x x · x x · x x = x^6

Three copies of x^2: 2 × 3 = 6. Power to a power → MULTIPLY.

Zero Exponent · $x^0 = 1$

See why → x x x ÷ x x x = 1, but by the quotient rule = x^{3-3} = x^0. So $x^0 = 1$

Anything (except 0) divided by itself is 1 → $x^0 = 1$.

Negative Exponent · $x^{-3} = 1/x^3$

See why → x x ÷ x x x x x = x^{2-5} = x^{-3}; cancelling leaves $1 / x^3$

A negative exponent means 'flip it' → $x^{-n} = 1/x^n$.

Quick Check

MC **Simplify $x^5 \cdot x^3$. Which rule applies, and what is the result?**

- ◯ Product Rule → x^8
- ◯ Power Rule → x^{15}
- ◯ Quotient Rule → x^2

Match the Rule to its Key Phrase

MATCH Draw a line from each rule to its key phrase.

Rule	Key Phrase
Product Rule	Power to power? MULTIPLY
Quotient Rule	Same base? ADD exponents
Power Rule	Flip to a fraction
Negative Exponent	Same base? SUBTRACT exponents

Worked Example — Simplify $(4x^3y^2)(3x^5y)$

Step 1 — Multiply coefficients: 4 × 3 = 12

Step 2 — Add x exponents: $x^3 \cdot x^5 = x^8$

Step 3 — Add y exponents: $y^2 \cdot y^1 = y^3$

✓ **Step 4 – Combine:** $12x^8y^3$

ERROR ANALYSIS **Spot the Error — Simplify $(3x^2)^3$.** A student wrote $3x^6$.

The coefficient 3 must ALSO be cubed: $3^3 = 27$. Correct: **$27x^6$**. The power rule applies to EVERYTHING inside the parentheses.

PRACTICE A Exponent Rules with Monomials

Rules: Multiply → ADD exponents | Divide → SUBTRACT | Power-to-Power → MULTIPLY

1. Simplify $x^3 \cdot x^5$ __________
2. Simplify $(y^4)^3$ __________
3. Simplify $a^8 \div a^3$ __________
4. Simplify $(2x^3)(5x^2)$ __________
5. Simplify $(3m^2)^3$ __________
6. Simplify $12y^7 \div 4y^2$ __________
7. Simplify $(-2a^4)^2$ __________
8. Simplify $(5x^2y)(3xy^3)$ __________
9. Simplify $18n^6 \div 6n^6$ __________
10. Simplify $(4b^3)^2 \div 2b$ __________
11. Simplify $(-3x^2)(-4x^3)$ __________
12. Simplify $(x^5y^2)^3$ __________

2 Scientific Notation

$$a \times 10^n \quad \text{where} \quad 1 \le a < 10$$

4.5×10^7 | coefficient · base (always 10) · exponent

SCIENTIFIC NOTATION: MOVE THE DECIMAL

$$45{,}000 \rightarrow 4.5 \times 10^4$$

big number → move LEFT 4 places → exponent +4

$$0.0067 \rightarrow 6.7 \times 10^{-3}$$

small number → move RIGHT 3 places → exponent −3

Type	Direction	Exponent	Example
Large number	Move decimal LEFT	______	$45{,}000 = 4.5 \times 10^4$
Small number (< 1)	Move decimal RIGHT	______	$0.0067 = 6.7 \times 10^{-3}$

PRACTICE B Scientific Notation Conversions

Remember: $a \times 10^n$ where $1 \le a < 10$. Positive exp = BIG number · Negative exp = SMALL number.

1. 45,000,000 → sci. notation __________________
2. 0.00032 → sci. notation __________________
3. 6.02×10^8 → standard __________________
4. 9.1×10^{-5} → standard __________________
5. 780,000 → sci. notation __________________
6. 0.0000067 → sci. notation __________________
7. 3.45×10^4 → standard __________________
8. 1.2×10^{-3} → standard __________________
9. 5,280,000,000 → sci. notation __________________
10. 2.5×10^{-2} → standard __________________

Multiplying Scientific Notation — 3-Step Process

1. __________ the coefficients (the a-values)
2. __________ the exponents
3. Adjust if coefficient is NOT between ______ and ______

Example (no adjustment): $(3 \times 10^4) \times (2 \times 10^5)$

Multiply coefficients: $3 \times 2 = 6$

ADD exponents: $4 + 5 = 9$

✓ **Answer:** 6×10^9

Dividing Scientific Notation — 3-Step Process

1. __________ the coefficients
2. __________ the exponents
3. Adjust if needed

Example (no adjustment): $(8 \times 10^6) \div (4 \times 10^2)$

Divide coefficients: $8 \div 4 = 2$

SUBTRACT exponents: $6 - 2 = 4$

✓ **Answer:** 2×10^4

⚠ **ADJUSTMENT RULES:**
- If coefficient ≥ 10: Move decimal LEFT, ADD 1 to exponent
- If coefficient < 1: Move decimal RIGHT, SUBTRACT 1 from exponent

Example needing adjustment: $(5 \times 10^3) \times (4 \times 10^2)$

Multiply: $5 \times 4 = 20$

Add exponents: $3 + 2 = 5$ → result: 20×10^5

Adjust (20 ≥ 10): $20 = 2.0 \times 10^1$, so ADD 1 to exponent

✓ **Final answer:** 2.0×10^6

ERROR ANALYSIS **Spot the Error — Convert 0.0045 to scientific notation.** A student wrote 4.5×10^3.
Small number → NEGATIVE exponent. Correct: $\mathbf{4.5 \times 10^{-3}}$. Numbers less than 1 always have negative exponents in scientific notation.

PRACTICE C Operations with Scientific Notation

Steps: Multiply/divide the coefficients · Add/subtract the exponents · Adjust if the coefficient is not between 1 and 10.

1. $(3 \times 10^4)(2 \times 10^5)$ ____________________
2. $(8 \times 10^7) \div (4 \times 10^3)$ ____________________
3. $(5 \times 10^3)(7 \times 10^6)$ ____________________
4. $(9.6 \times 10^8) \div (3.2 \times 10^5)$ ____________________

Word Problems with Scientific Notation

Strategy: Identify what operation to use. Write your scientific-notation setup. Adjust if needed.

PRACTICE D Word Problems

1. The Sun is about 9.3×10^7 miles from Earth. Light travels 1.86×10^5 miles per second. How many seconds does it take light to reach Earth?

Answer: ____________________

2. A bacteria cell is about 2×10^{-6} m long. A virus is about 5×10^{-8} m long. How many times longer is the bacteria?

Answer: ____________________

3. The US national debt is about 3.4×10^{13} dollars. The US population is about 3.4×10^{8} people. What is the debt per person?

Answer: ____________________

4. A grain of sand has mass 6.5×10^{-4} grams. A beach has about 8×10^{18} grains of sand. What is the total mass in grams?

Answer: ____________________

5. There are about 3.7×10^{13} cells in a human body. Each cell is about 1×10^{-5} m wide. If all cells were lined up, how long in meters?

Answer: ____________________

6. A computer does 4.8×10^{9} calculations per second. A task needs 1.44×10^{12} calculations. How many seconds to finish?

Answer: ____________________

★ Cumulative Practice — Unit 2 Review ★

Mixed practice over everything above. Show all work; circle your final answer.

REVIEW · A Exponent Rules

1. Simplify $x^4 \cdot x^6$ ____________
2. Simplify $(m^3)^4$ ____________
3. Simplify $p^9 \div p^4$ ____________
4. Simplify $(3a^2)(4a^5)$ ____________
5. Simplify $(-2c^4)^3$ ____________
6. Simplify $(6x^5y^3) \div (2x^2y)$ ____________

REVIEW · B Scientific Notation Conversions

1. 6,300,000 → sci. notation ____________
2. 0.00071 → sci. notation ____________
3. 2.08×10^5 → standard ____________
4. 4.5×10^{-6} → standard ____________
5. 91,000,000 → sci. notation ____________
6. 3.6×10^{-2} → standard ____________

REVIEW · C Operations

1. $(2 \times 10^3)(4 \times 10^6)$ ____________
2. $(9 \times 10^8) \div (3 \times 10^2)$ ____________
3. $(6 \times 10^5)(5 \times 10^4)$ ____________
4. $(7.2 \times 10^7) \div (1.8 \times 10^3)$ ____________

REVIEW · D Word Problems

1. A red blood cell is about 8×10^{-6} m wide. A platelet is about 2×10^{-6} m wide. How many times wider is the red blood cell?

Answer: ____________

2. Earth is about 1.5×10^8 km from the Sun. A spacecraft travels 3×10^4 km per hour. How many hours would the trip take?

Answer: ____________

3. A factory makes 5×10^3 widgets per day. How many widgets does it make in 2×10^2 days?

Answer: ____________

★ Answer Key ★

Do not distribute to students until after the lesson.

Guided Notes — Fill-Ins

2.1 Exponent Rules. Product: $x^3 \times x^4 = x^7$; $(2x^3)(3x^2) = 6x^5$; "Same base? ADD." **Quotient:** $x^6 \div x^2 = x^4$; $(12x^7) \div (3x^2) = 4x^5$; "Same base? SUBTRACT." **Power:** $(x^2)^3 = x^6$; $(2x^3)^2 = 4x^6$; "Power to power? MULTIPLY." **Zero:** $5^0 = 1$; $(3x^5)^0 = 1$; "ANY $base^0 = 1$." **Negative:** $2^{-3} = 1/8$; $4x^{-3} = 4/x^3$; "Flip to a fraction." Parentheses: $-3^2 = -9$ vs $(-3)^2 = 9$.

Quick Check: Product Rule → $x^5 \cdot x^3 = x^8$. **Match:** Product→"Same base? ADD"; Quotient→"Same base? SUBTRACT"; Power→"Power to power? MULTIPLY"; Negative→"Flip to a fraction." **Worked:** $(4x^3y^2)(3x^5y) = 12x^8y^3$. **Error:** $(3x^2)^3 = 27x^6$.

2.2 Scientific Notation. Table: Large number → exponent **+** (positive); Small number → exponent **−** (negative). **Multiply:** MULTIPLY coefficients, ADD exponents, adjust to between 1 and 10. **Divide:** DIVIDE coefficients, SUBTRACT exponents. Examples: $(3\times10^4)(2\times10^5) = 6\times10^9$; $(8\times10^6) \div (4\times10^2) = 2\times10^4$; $(5\times10^3)(4\times10^2) = 20\times10^5 \rightarrow 2.0\times10^6$. **Error:** $0.0045 = 4.5\times10^{-3}$.

Practice A — Exponent Rules with Monomials

1) x^8 2) y^{12} 3) a^5 4) $10x^5$ 5) $27m^6$ 6) $3y^5$ 7) $4a^8$ 8) $15x^3y^4$ 9) 3 10) $8b^5$ 11) $12x^5$ 12) $x^{15}y^6$

Practice B — Conversions

1) 4.5×10^7 2) 3.2×10^{-4} 3) 602,000,000 4) 0.000091 5) 7.8×10^5 6) 6.7×10^{-6} 7) 34,500 8) 0.0012 9) 5.28×10^9 10) 0.025

Practice C — Operations

1) 6×10^9 2) 2×10^4 3) 3.5×10^{10} 4) 3×10^3

Practice D — Word Problems

1) $(9.3\times10^7) \div (1.86\times10^5) = 5\times10^2$ s (500 s) 2) $(2\times10^{-6}) \div (5\times10^{-8}) = 40\times$ longer 3) $(3.4\times10^{13}) \div (3.4\times10^8) = 1\times10^5$ = \$100,000 4) $(6.5\times10^{-4})(8\times10^{18}) = 5.2\times10^{15}$ g 5) $(3.7\times10^{13})(1\times10^{-5}) = 3.7\times10^8$ m 6) $(1.44\times10^{12}) \div (4.8\times10^9) = 3\times10^2$ s (300 s)

Cumulative Review

A · Exponent Rules: 1) x^{10} 2) m^{12} 3) p^5 4) $12a^7$ 5) $-8c^{12}$ 6) $3x^3y^2$

B · Conversions: 1) 6.3×10^6 2) 7.1×10^{-4} 3) 208,000 4) 0.0000045 5) 9.1×10^7 6) 0.036

C · Operations: 1) 8×10^9 2) 3×10^6 3) $30\times10^9 \rightarrow 3.0\times10^{10}$ 4) 4×10^4

D · Word Problems: 1) $(8\times10^{-6}) \div (2\times10^{-6}) = 4\times$ wider 2) $(1.5\times10^8) \div (3\times10^4) = 0.5\times10^4 = 5\times10^3$ h (5,000 h) 3) $(5\times10^3)(2\times10^2) = 10\times10^5 = 1\times10^6$ widgets